

Asynchronous Activity 1 Evidence

KNIME Student Early Warning Tool and GitHub Submission

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Course and section: CS0065 - Intelligent Systems, AN43

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Evidence order

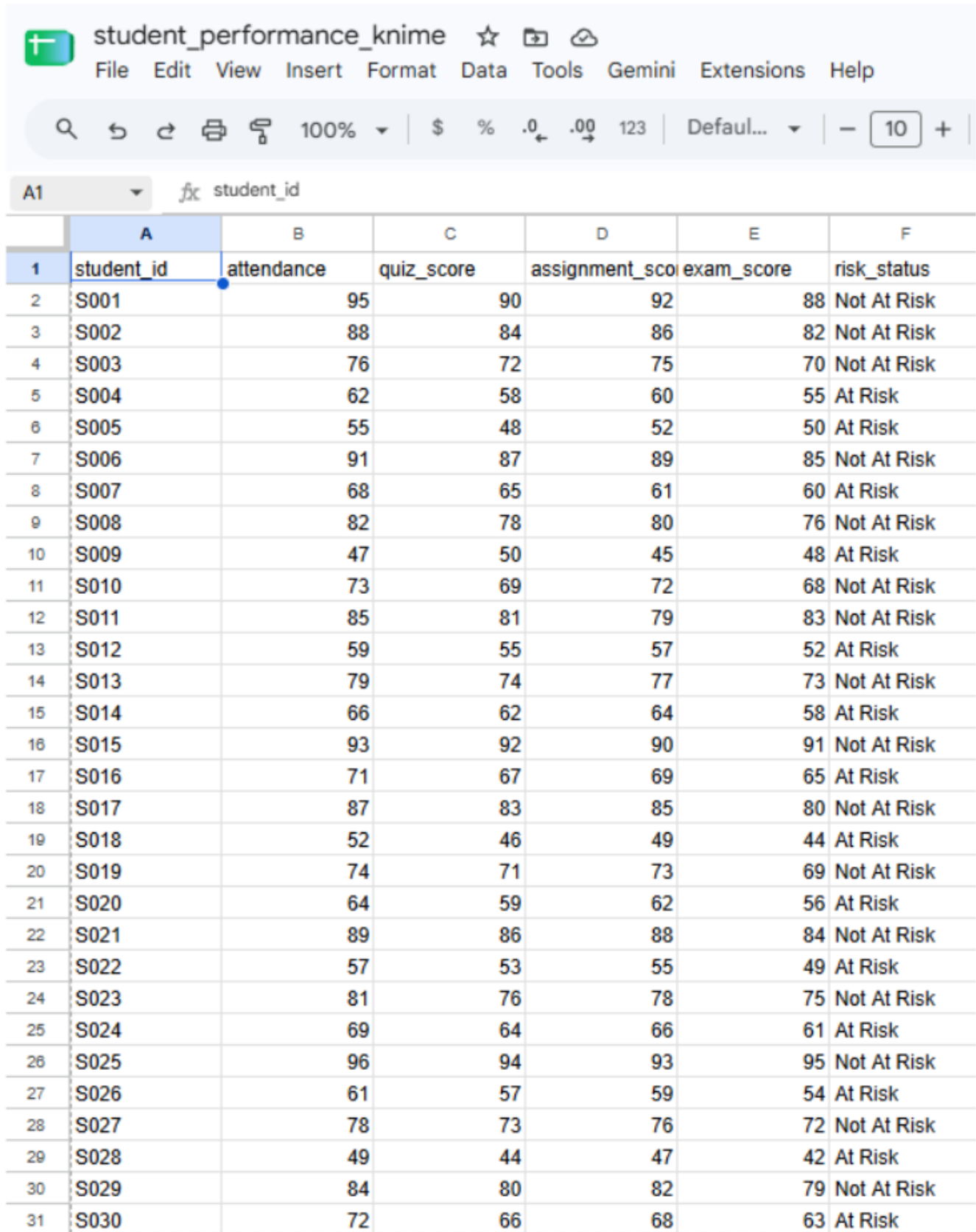
1. Dataset and input configuration
2. Column selection and train/test partition
3. Decision Tree configuration and evaluation
4. Completed classifier workflow
5. GitHub repository verification

Screenshots are arranged in the order of the activity instructions.

Step 1 - Dataset prepared with the required student performance columns and risk labels.

Gorospe Paolo C

1.



student_performance_knime						
File Edit View Insert Format Data Tools Gemini Extensions Help						
100% \$ % .0 .00 123 Default... - 10 +						
A1	fx student_id					
	A	B	C	D	E	F
1	student_id	attendance	quiz_score	assignment_score	exam_score	risk_status
2	S001	95	90	92	88	Not At Risk
3	S002	88	84	86	82	Not At Risk
4	S003	76	72	75	70	Not At Risk
5	S004	62	58	60	55	At Risk
6	S005	55	48	52	50	At Risk
7	S006	91	87	89	85	Not At Risk
8	S007	68	65	61	60	At Risk
9	S008	82	78	80	76	Not At Risk
10	S009	47	50	45	48	At Risk
11	S010	73	69	72	68	Not At Risk
12	S011	85	81	79	83	Not At Risk
13	S012	59	55	57	52	At Risk
14	S013	79	74	77	73	Not At Risk
15	S014	66	62	64	58	At Risk
16	S015	93	92	90	91	Not At Risk
17	S016	71	67	69	65	At Risk
18	S017	87	83	85	80	Not At Risk
19	S018	52	46	49	44	At Risk
20	S019	74	71	73	69	Not At Risk
21	S020	64	59	62	56	At Risk
22	S021	89	86	88	84	Not At Risk
23	S022	57	53	55	49	At Risk
24	S023	81	76	78	75	Not At Risk
25	S024	69	64	66	61	At Risk
26	S025	96	94	93	95	Not At Risk
27	S026	61	57	59	54	At Risk
28	S027	78	73	76	72	Not At Risk
29	S028	49	44	47	42	At Risk
30	S029	84	80	82	79	Not At Risk
31	S030	72	66	68	63	At Risk

2.

Step 2 - CSV Reader configured to read the student performance dataset.

CSV Reader

Add comment

Column Filter

Table Partitioner

Decision

84%

CSV Reader

File

Mode

FileFiles in folders

Source

C:\Users\paolo\Downloads\student_performance_knim...

File Format

Skip first lines of file

0

Comment line character

#

Row delimiter

Line breakCustom

Column delimiter

.

☐ Quoted strings contain no row delimiters

Quote character

"

Quote escape character

"

☒ Autodetect format

Data Area

☒ First row contains column names

Skip first data rows

0

Discard

Apply

Apply and Execute

1: File Table

Flow Variables

Rows: 30 | Columns: 6

Sorting

Filter

Table

Statistics

Search

#	ROWID	student_id	attendance	quiz_score	assignment_s...	exam_score	risk_status
		String	Number (Integer)	Number (Integer)	Number (Integer)	Number (Integer)	String
1	Row0	S001	95	90	92	88	Not At Risk
2	Row1	S002	88	84	86	82	Not At Risk
3	Row2	S003	76	72	75	70	Not At Risk
4	Row3	S004	62	58	60	55	At Risk
5	Row4	S005	55	48	52	50	At Risk
6	Row5	S006	91	87	89	85	Not At Risk
7	Row6	S007	68	65	61	60	At Risk
8	Row7	S008	82	78	80	76	Not At Risk
9	Row8	S009	47	50	45	48	At Risk
10	Row9	S010	73	69	72	68	Not At Risk
11	Row10	S011	85	81	79	83	Not At Risk
12	Row11	S012	59	55	57	52	At Risk
13	Row12	S013	79	74	77	73	Not At Risk

Step 3 - Column Filter keeps the four features and the risk_status target.

3.

CSV Reader

Column Filter

Table Partitioner

Decision Leaf

1: Filtered table

Flow Variables

Column Filter

Column filter

Manual Pattern Type

Search

Excludes

Includes

student_id

attendance

quiz_score

assignment_score

exam_score

risk_status

Any unknown column

Discard

Apply

Apply and Execute

Step 4 - Table Partitioner configured for a 70/30 stratified split with seed 42.

4.

CSV Reader

Column Filter

Table Partitioner

Add comment

Decision Leaf

- 84% +

Table Partitioner

First partition type

Relative (%)

Absolute

Relative size

70

Sampling strategy

Random

Stratified

Linear

First rows

Group column

risk_status

☒ Fixed random seed

42

If input table is empty

Fail

Output empty table(s)

Discard

Apply

Apply and Execute

Step 5 - Decision Tree Learner configured with risk_status as the class column.

5.

The diagram illustrates a data processing workflow. It begins with a 'Column Filter' node, which connects to a 'Table Partitioner' node. The 'Table Partitioner' node then branches into two paths: one leading to a 'Decision Tree Learner' node and another leading to a 'Decision Tree Predictor' node. The 'Decision Tree Learner' node is highlighted with a blue box and has a tooltip that says 'Add comment'. The 'Decision Tree Predictor' node is highlighted with a green box. Below the workflow, there is a small preview window showing a scatter plot with a blue rectangle and several colored dots (orange, yellow, green, red). At the bottom of the workflow area, there are icons for a hand, a book, and a zoom control set to 84%.

Decision Tree Learner ×

Class column

risk_status

Quality measure

Gini index

Gain ratio

Pruning method

No pruning

MDL

☒ Reduced error pruning

Minimum number of records per node

2

Number of records to store for view

10000

☒ Average split point

Number threads

12

☒ Skip nominal columns without domain information
☐ Force root split column
☐ Binary nominal splits

PMML Settings
No true child strategy

Return null prediction

Return last prediction

Missing value strategy

Last prediction

None

Discard

Apply

Apply and Execute

■ 1: Decision Tree Model

☑ Flow Variables

Step 6 - Decision Tree Predictor connected to the trained model and testing data.

6.

The workflow diagram illustrates the process of training and testing a decision tree model. It begins with a 'Column Filter' node, which feeds into a 'Table Partitioner' node. The 'Table Partitioner' node then splits the data into two paths: one leading to a 'Decision Tree Learner' node and another leading to a 'Decision Tree Predictor' node. The 'Decision Tree Learner' node outputs a model, which is then used by the 'Decision Tree Predictor' node to make predictions on the testing data. The 'Decision Tree Predictor' node is highlighted with a blue border, indicating it is the current step in the workflow.

Decision Tree Predictor

Number of patterns for hiliting

10000

☐ Custom prediction column name

☐ Append columns with normalized class distribution

Discard

Apply

Apply and Execute

Step 7 - Decision Tree Scorer confusion matrix configured with the actual and predicted columns.

7.

Decision Tree Learner

Decision Tree Predictor

Scorer

Table Partitioner

Scorer

First column

T risk_status

Second column

T Prediction (risk_status)

Sorting strategy

Insertion order

Missing values

Ignore

Fail

Discard

Apply

Apply and Execute

1: Confusion matrix

2: Accuracy statistics

Flow Variables

Rows: 2 | Columns: 2

Sorting

Filter

Table

Statistics

Search

#	RowID	Not At Risk	At Risk
		Number (integer)	Number (integer)
1	Not A...	5	0
2	At Risk	0	4

Step 8 - Decision Tree Scorer accuracy statistics displayed.

8.

Decision Tree
Learner

Decision Tree
Predictor

Scorer

ble Partitioner

84%

Scorer

First column

risk_status

Second column

Prediction (risk_status)

Sorting strategy

Insertion order

Reverse order

Use name prefix

Missing values

Ignore

Fail

Discard

Apply

Apply and Execute

1: Confusion matrix

2: Accuracy statistics

Flow Variables

Rows: 3 | Columns: 11

Sorting

Filter

Table

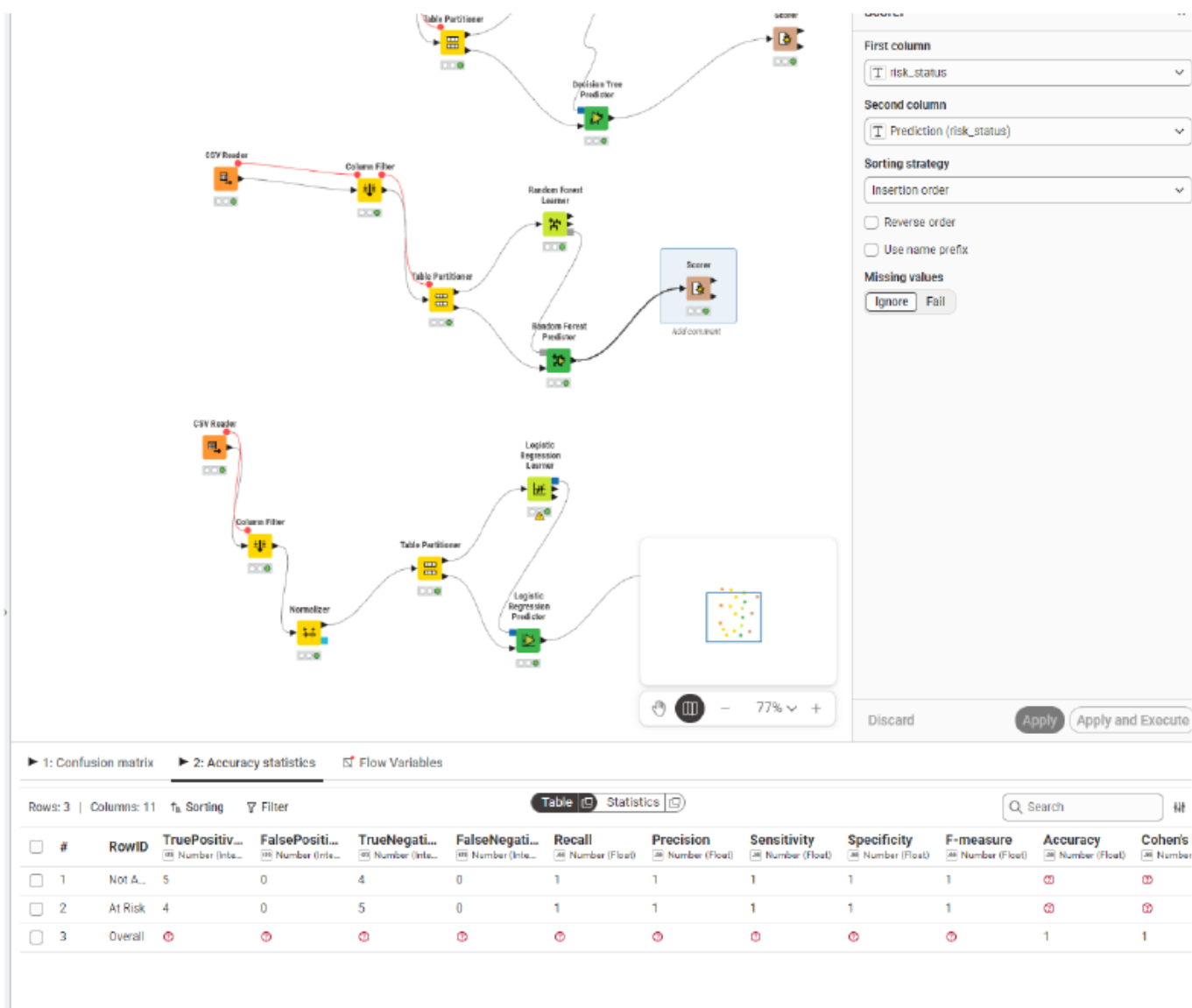
Statistics

Search

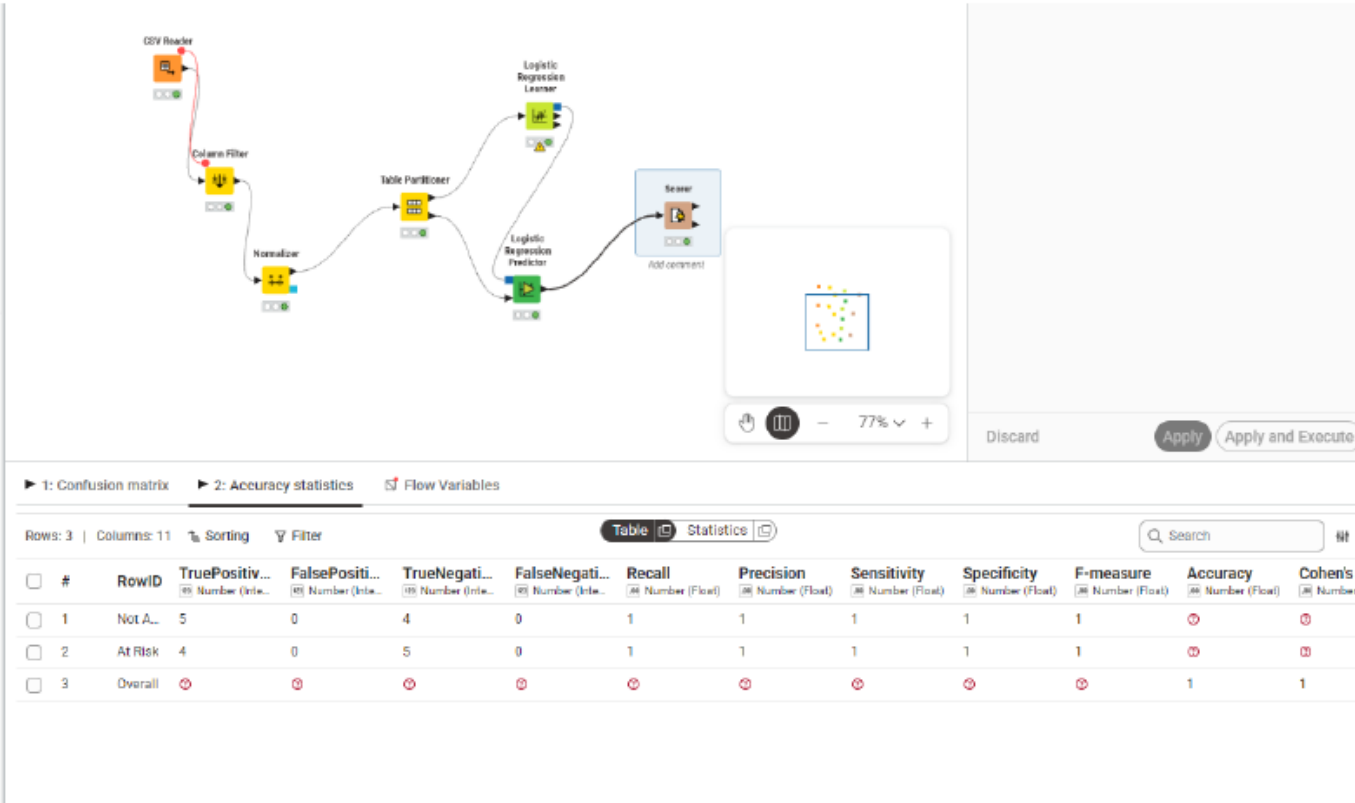
#	RowID	TruePositiv...	FalsePositi...	TrueNegati...	FalseNegati...	Recall	Precision	Sensitivity
		Number (Inte...	Number (Inte...	Number (Inte...	Number (Inte...	Number (Float)	Number (Float)	Number (Float)
1	Not A...	5	0	4	0	1	1	1
2	At Risk	4	0	5	0	1	1	1
3	Overall							

9.

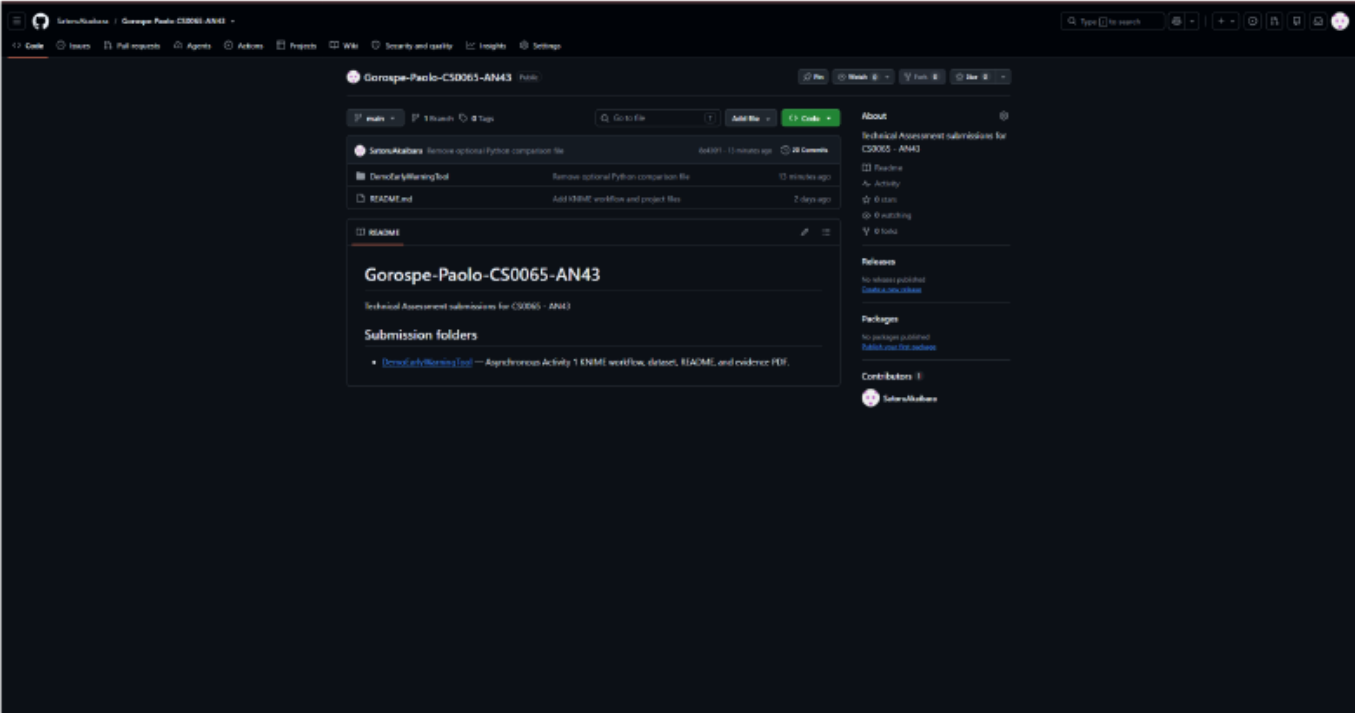
Step 9 - Completed KNIME workflow showing the classifier branches and evaluation output.



Step 10 - Public GitHub repository root showing the project repository.



11.



12.

Step 11 - Public GitHub DemoEarlyWarningTool folder showing the project files.

